Player Churn Prediction

Technical Report: Baseline Model and End-to-End Pipeline

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Abstract

I predict, for each active player of a brand, the chance of 60 days without an active day (a day with a stake from real money or released bonus). One command (`make pipeline AS_OF=date`) reads the S3 data lake (`org/40-gold`), builds 39 features, trains a calibrated LightGBM and a Cox proportional hazards (Cox PH) model, runs an 11 month backtest and writes the daily scores. In the backtest the mean AUC is 0.87 to 0.89 in five brands, 0.10 to 0.12 above recency alone. A campaign on the 1,000 players with the highest value at risk reaches 59% to 80% of the gross gaming revenue (GGR) at risk, against 8% to 28% with recency; most of this gain comes from using value, and the model adds 6 to 7 points over recency weighted by value. The data go from 2025-04-01 to 2026-10-06.

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1 Summary

Table 1 gives the main results. All of them use the official active day: the backtests ran on 15 October 2026 and the registered models were trained again on 16 October 2026. The example figures are for brand 64; the result tables cover the five brands.

Table 1: Main results.

Topic	Result	Section
Churn	60 days without an active day. With 30 days, 21% to 29% of the churned players come back.	2.3
Active day	A stake from real money or released bonus, as in the variable definitions of the gold data model. Playable bonus alone does not count; this removes 18% to 44% of the players I first counted as active.	2.2
Models	Calibrated LightGBM for the 60-day probability; Cox PH for the 7, 14 and 30-day horizons.	3.4
Ranking	Backtest AUC 0.87 to 0.89, 0.10 to 0.12 above recency; the model beats recency in all 55 brand-months and is never below 0.83.	4.2
Features	Frequency (`active_days_l90d`) is the strongest family, then recency and tenure.	4.1
Calibration	The only failure of the backtest: ECE above 0.08 in 5 brand-months, four of them in December 2025. Brand 73 passes every condition.	4.5
Money	With the same 1,000 contacts, ranking by value at risk reaches 3 to 8 times more historical GGR opportunity than value-blind recency. Most of it comes from using value; the model adds 6 to 7 points over recency weighted by value. It is an opportunity, not GGR saved.	4.3
Handoff score	Since 16 October 2026 `value_at_risk` uses LightGBM at 60 days instead of Cox at 30 days: with both calibrated, LightGBM reaches more GGR at risk in every brand-month tested (3.5 points on average).	4.3
Source data	Recency, tenure and active days are wrong in the signals table; the pipeline computes them from the daily activity.	5
Plan	Of the 40 Phase 1 items, 25 are done, 14 changed with a reason and 1 deferred.	6

2 Data and definitions

2.1 Data

Table 2 lists the tables I use and their dates. S3 access is in `src/features/datalake.py` (DuckDB on Parquet), and `src/features/daily_cache.py` keeps local daily copies, so S3 is not read on every run. The data-quality profile uses SQL aggregations on the full tables, with no sampling (`eda/profile_tables.py`, reports in `docs/dq_reports/`).

Table 2: Data lake tables used in the project.

Table	Rows	Dates	Use
`gld_player_gaming_daily`	59.6 M	2025-04-01 to 2026-10-06	Days with a bet, by wallet: the base of the active days, churn and recency
`gld_player_financial_daily`	688 M	2025-04-01 to 2026-10-06	Wagered, won, GGR and bonus per day
`gld_player_payments_daily`	5.6 M	2025-09-18 to 2026-10-06	Deposits and withdrawals; 40 days missing; completed deposits only from March 2026
`gld_player_signals_daily`	689 M	2025-04-01 to 2026-10-06	Precomputed signals; `brand_id` is empty before July 2026

Limits. Completed deposits only exist from March 2026, so before that the deposit features are empty, not 0. The history was computed again in a backfill that ended on 2026-08-08. There are only 18 months, with no full previous year to model seasonality.

2.2 Active day

An active day is a day on which the player staked a non-zero amount from real money or released bonus funds (this is how the variable definitions of the gold data model define it). Stakes only from playable bonus do not count: these funds still have a wagering requirement and

cannot be withdrawn until the bonus conditions are met. For player i and day d:

$$a_{i,d} = \begin{cases} 1 & \text{if } W^{\text{real}}_{i,d} + W^{\text{released}}_{i,d} > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

where $W^{\text{real}}_{i,d}$ and $W^{\text{released}}_{i,d}$ are the amounts staked that day from each wallet (`wagered_real` and `wagered_released_bonus` of `gld_player_gaming_daily`, added over games, providers and currencies).

Use. The activity cache keeps every bet, because bet counts and amounts wagered include all wallets, and marks each day with `active`. The population, the churn and survival labels, recency, tenure, the `active_days_*` features and the early churn estimate (Section 4.5) read the active days only. A test (`test_active_day_definition.py`) covers real money only, released bonus only, playable bonus only, mixed wallets and empty amounts, and a cache built with an older definition is refused. No day marked as not active has a real-money stake in `gld_player_financial_daily` (0.00% in the five brands); the 8% to 15% of active days with no real-money stake there are days with released bonus only.

Effect. Until 15 October 2026 the pipeline counted a bet from any wallet. The official definition mainly changes the population (Table 3): players who only play with playable bonus leave often, so without them the churn rate is lower. For the players who stay, only 2% to 5% of the labels change.

Table 3: Effect of the official active day (Equation (1)) against a bet from any wallet. Means over the 12 monthly cutoffs from September 2025 to August 2026, except the first column.

Brand	Bet days not active (range by month)	Players who leave the population	60-day churn rate before → after	Labels that change
14	6% to 21%	44%	55.6% → 34.1%	2.1%
23	10% to 32%	24%	29.3% → 25.2%	5.0%
64	5% to 24%	24%	40.1% → 32.7%	2.9%
72	13% to 34%	35%	59.5% → 51.6%	3.0%
73	8% to 24%	18%	43.2% → 40.5%	3.0%

Note. Bet days not active: share of the player days with a bet that have stakes from playable bonus only, from April 2025 to September 2026. Labels that change: share of the players in both populations whose 60-day label is different. Source: activity cache, aggregates only.

2.3 Churn label and cutoffs

A player churns after 60 days without an active day. Each cutoff c is the first day of a month. Its population is the players with an active day in $(c-30, c]$, and its label looks at the next 60 days (Figure 1):

$$y_{i,c} = \begin{cases} 1 & \text{if player } i \text{ has no active day in } (c, c+60], \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Code: `src/features/churn_labels.py`; analysis: `eda/02_churn_definition.ipynb`.

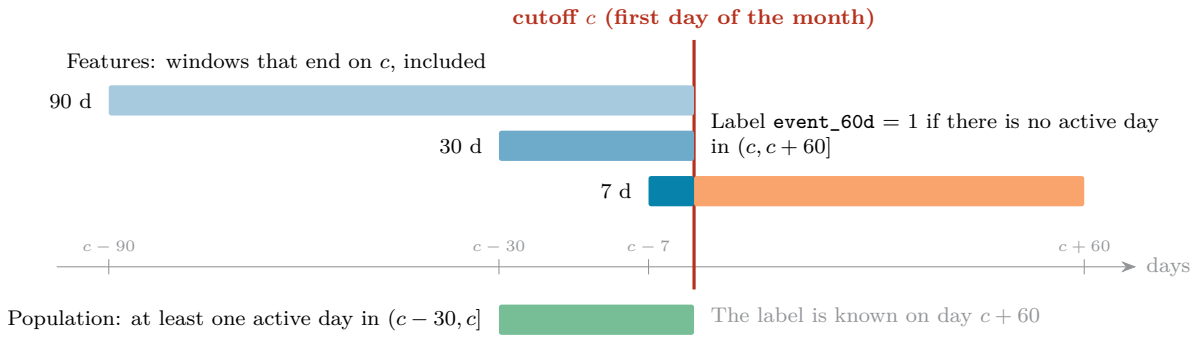


Figure 1: How a cutoff c is defined: the features look back, the label looks forward.

Why 60 days (Table 4). With 30 days too many churned players come back. A label of N days is known N days after the cutoff, so 90 days would delay every label by one more month, leave the

calibration further behind the season (Section 4.5) and lose one cutoff, for 4 to 6 points less false churn. With the official active day the choice is closer than before: playing only with playable bonus now counts as silence, and many of these players come back.

Table 4: False churn by threshold: players who have an active day again in the next 60 days.

Silent days	Brand 64	All brands	Reading
30	29.1%	21.4%	Too much false churn
60	**16.7%**	**12.3%**	Below 15% for all brands together, but not in brands 14, 23 and 64 (17% to 19%)
90	11.0%	8.2%	4 to 6 points lower, for one more month of waiting

Survival labels (`docs/survival_data_card.md`). `event_60d` is $y_{i,c}$, the LightGBM target. `duration_days` and `event_observed` are the Cox PH target, with right censoring (Kaplan and Meier, 1958; Cox, 1972), and `churn_within_7d/14d/30d` calibrate the Cox horizons. Time counts from the cutoff and every player enters on day 0, so there is no left truncation. The only censoring is the data end (15% to 41% by brand, more in later cutoffs). Every cutoff has at least 60 days of follow-up, so the 30-day horizon is never censored, and `event_60d` $= 1$ always means an event on day 0, so the labels agree. Tenure is capped at 150 days and means the same only from 29 August 2025, so in the first two training months of the backtest older players look younger.

3 Method

3.1 Pipeline and daily scoring

`src/features/run_pipeline.py` runs every step as of `AS_OF`, with no later data. Each step uses the exact file or MLflow run of the step before (Figure 2), and everything is logged in MLflow (`make mlflow-ui`). The last step, `src/models/score.py`, builds the features with the training code, runs the alerts, estimates the calibration level again and writes `data/03_output/player_scores/run_date=.../brand_id=N.parquet`; running a date again overwrites it. Main columns: `p_churn_60d`, `p_churn_7d/14d/30d`, `value_at_risk`, `risk_band` (decile), `top_drivers` (TreeSHAP) and `drift_flag` (`docs/schema_player_scores.md`). The model card (`src/models/model_card.py`) sums up each model in one HTML page attached to its MLflow version.

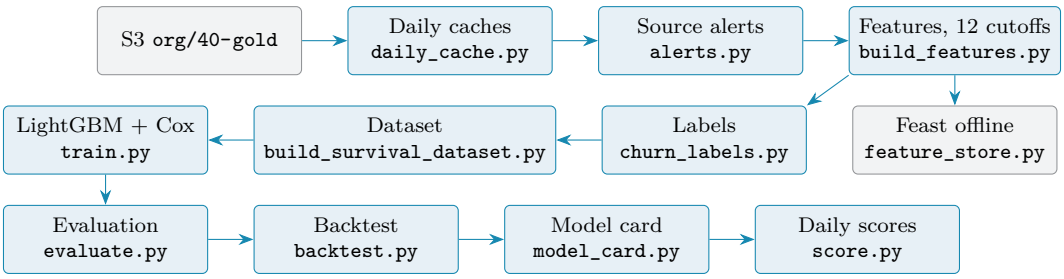


Figure 2: The flow of `make pipeline`. Each box shows the file that runs the step.

3.2 Features

I tested 32 signal hypotheses on 12 training cutoffs with four measures (lift, univariate Cox hazard ratio, single-signal AUC and stability) and promoted 21 (`eda/signal_explorer.py`). The result is 39 features in 8 families (Table 5, dictionary in `docs/feature_dictionary.md`), built with data up to the cutoff (`src/features/build_features.py`). Amounts are capped at the 99.5th percentile $q_{0.995}$ of each brand, learned on the training cutoffs only (`winsorisation.py`), and scaled with a signed log:

$$\tilde{x} = \operatorname{sign}(x)\,\log\bigl(1 + |x|\bigr), \qquad x = \min\bigl(x_{\text{raw}},\, q_{0.995}\bigr). \tag{3}$$

A test (`test_leakage.py`) checks that no feature uses data after the cutoff. Feast (`src/features/feature_repo/definitions.py`) writes the offline store and `feature_store.py` checks the point-in-time join; it is local because I only have read access to AWS.

Table 5: Feature families.

Family	N	Examples
Recency	2	`days_since_last_bet`, `days_since_last_deposit`
Frequency	9	`active_days_l7d/l30d/l90d`, `bets_l30d`, `sessions_l7d`
Monetary	7	`wagered_eur_l7d`, `ggr_eur_l30d`, `deposits_eur_l30d`
Lag	3	`prior_wagered_eur_l7d`, `prior_active_days_l7d`
Trend	3	`wagered_trend_7`, `heavy_loss_multiple`
Tenure and life cycle	3	`days_since_first_bet`, `prior_dormancy_spells_14d`
Mix and behaviour	7	`engagement_score`, `games_breadth_30d`, `losing_streak`
Deposits	5	`deposited_within_3d/7d/14d`, `failed_deposits_14d`

3.3 Data quality and alerts

The data-quality checks run in four stages (Table 6).

Table 6: The four stages of the data-quality checks (limits in `configs/eda_alerts.yaml`).

Stage	What it checks	File
1 Quality	Pandera suites for each cache and for the features	`eda/expectations/`
2 Anomalies	IQR, MAD, Isolation Forest and daily totals; nothing is deleted, values are capped	`eda/anomalies.py`
3 Alerts	Rows, nulls, totals and PSI. A critical alert stops the pipeline, unless a person accepted it for a limited time; drift only sets `drift_flag`	`alerts.py`
4 Summary	Each statistic against the previous dataset of the brand	`eda/summary.py`

3.4 Models and metrics

All the training is in `src/models/train.py` (Table 7); the models and their parameters are in `catalog/model_library_catalog.json`. The Cox PH model (Cox, 1972), fitted with lifelines (Davidson-Pilon, 2019), uses the same features, is stratified by brand and calibrates its 7, 14 and 30-day horizons. The evaluation (`src/evaluation/evaluate.py`) gives AUC, ECE, log-loss and top-decile precision for LightGBM, and the C-index (Harrell et al., 1982), the integrated Brier score (IBS) and the time-dependent AUC for Cox, all with 95% bootstrap intervals. The ECE uses $B = 10$ bins of equal size:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \left| \bar{y}_b - \bar{p}_b \right|, \tag{4}$$

where n_b, $\bar{y}_b$ and $\bar{p}_b$ are the players, the observed churn rate and the mean predicted probability of bin b.

Table 7: LightGBM training steps. Each step is a nested MLflow run.

Step	What it does
Selection	Drops features with more than 40% nulls, almost constant or highly correlated (over 0.95); adversarial validation with a LightGBM; pruning by permutation importance (`select_features.py`)
Reference	LightGBM with default values: 63 leaves, learning rate 0.05, early stopping
Imbalance	`scale_pos_weight`, `is_unbalance` or nothing, chosen by log-loss after calibration
Optuna	100 trials (20 in the backtest) that minimise the validation log-loss
Regularisation	Only if the training AUC is more than 0.03 above the validation AUC
Calibration	Isotonic or Platt per brand, chosen by cross-validated ECE (`calibration.py`)

3.5 Backtest design

The registered model uses one time split: months 1 to 9 for training, month 10 for validation and months 11 and 12 for test. The rolling-origin backtest (`src/models/backtest.py`) trains again for each test month with only the cutoffs whose label was closed on that day; the month just before the test is never closed, so it is left out (Figure 3).

11 test months, not 12. The configuration asks for 12 (`configs/backtest.yaml`), but the history is still too short. The data start on 1 April 2025 and the longest feature window is 90 days, so the first cutoff is 1 July 2025. A test month needs two earlier months with a closed 60-day label (one to train and one to validate), so the first possible test month is October 2025. On the run date (6 October 2026) the last closed label is the one of 1 August 2026, which gives 11 test months, October 2025 to August 2026. **The label of 1 September 2026 closes on 31 October 2026, so a backtest run on that day or later has 12 test months** (October 2025 to September 2026), with no change to the code. All test months are first days of the month; other scoring days are not checked yet.

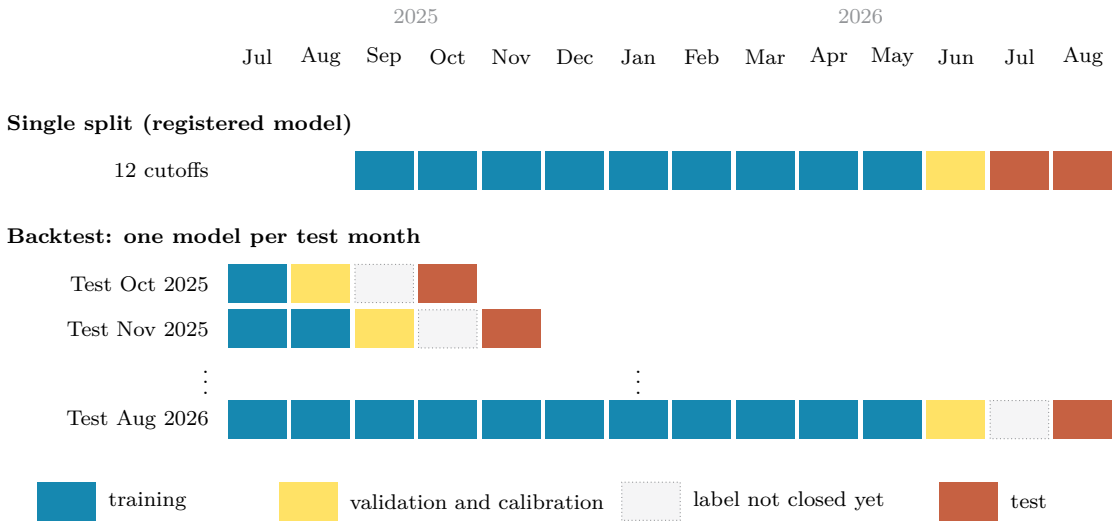


Figure 3: Training, validation and test (cutoffs of brand 73). In the backtest, the training data grows from 2 to 12 months.

Comparators. The optimised and the reference LightGBM (what tuning adds), Cox PH and the recency rule, the baseline that the model must beat. Recency is not a trained model: its score is the number of days since the last active day,

$$s_{i,c}^{\text{rec}} = c - t_{i,c}^{\text{last}}, \tag{5}$$

taken from the daily activity, not from the broken column of the signals table. For the value policies I also turn it into a probability with an isotonic regression on the training months (*calibrated recency*).

Campaign replay. At each cutoff the 1,000 players with the highest score are contacted (a provisional size, until the CRM team confirms it), and I add up the *GGR at risk reached*: the expected 30-day GGR of the contacted players who really churned,

$$G_i = \max\Big(0,\ \frac{30}{90}\,\text{GGR}_i^{(90)}\Big), \tag{6}$$

where $\text{GGR}_i^{(90)}$ is the GGR of the last 90 days. LightGBM ranks by the value at risk $G_i\,p_i^{(60)}$, with its calibrated 60-day probability; the daily scores hand off the same value at risk (`value_at_risk`). Until 16 October 2026 they multiplied G_i by the Cox probability of churn within 30 days (`value_at_risk_30d`).

Pass conditions. AUC ≥ 0.70 and ECE ≤ 0.08 in every month, and a top-decile ratio $R \geq 1.10$:

$$R = \frac{\overline{\text{Prec}}_{10\%}(\text{model})}{\overline{\text{Prec}}_{10\%}(\text{recency})}, \tag{7}$$

the mean precision of the model in the riskiest 10% of players divided by the same value for recency.

4 Results

4.1 Registered models

Table 8: Registered model of each brand, test in July and August 2026.

Brand	Version	Selected features	AUC	ECE
14	v2	`active_days_l90d`, `wagered_eur_l7d`, `days_since_first_bet`, `days_since_last_bet`	0.896	0.010
23	v2	`active_days_l30d`, `prior_dormancy_spells_14d`, `days_since_last_bet`, `prior_ggr_eur_l30d`, `bets_l7d`, `prior_wagered_eur_l7d`, `losing_streak`	0.893	0.024
64	v64	`active_days_l90d`, `days_since_last_bet`, `prior_dormancy_spells_14d`, `wagered_eur_l7d`, `engagement_score`	0.888	0.031
72	v2	`active_days_l90d`, `days_since_last_bet`, `prior_dormancy_spells_14d`, `wagered_eur_l7d`, `prior_ggr_eur_l30d`, `prior_wagered_eur_l7d`	0.878	0.016
73	v2	`active_days_l90d`, `days_since_last_bet`, `prior_dormancy_spells_14d`, `days_since_last_return`, `sessions_l7d`, `wagered_eur_l7d`	0.890	0.012

Note. AUC: area under the ROC curve. ECE: expected calibration error, Equation (4). Test churn rate: 18% (brand 23) to 44% (brand 72). Cox PH of brand 64: C-index 0.855 [0.851, 0.860], IBS 0.039. Source: MLflow runs of each registered version.

Table 8 gives the registered model of each brand. The models use 4 to 7 features; the 90-day frequency is in four brands, and brand 23 uses the 30-day one. Tuning adds nothing: in the backtest, the reference and the optimised configurations are at most 0.001 of AUC apart. For brand 64 (Figure 4), the training, validation and test AUC are 0.878, 0.884 and 0.888, so I see no material gap between them, and on the test months the model is far above recency (0.887 against 0.732).

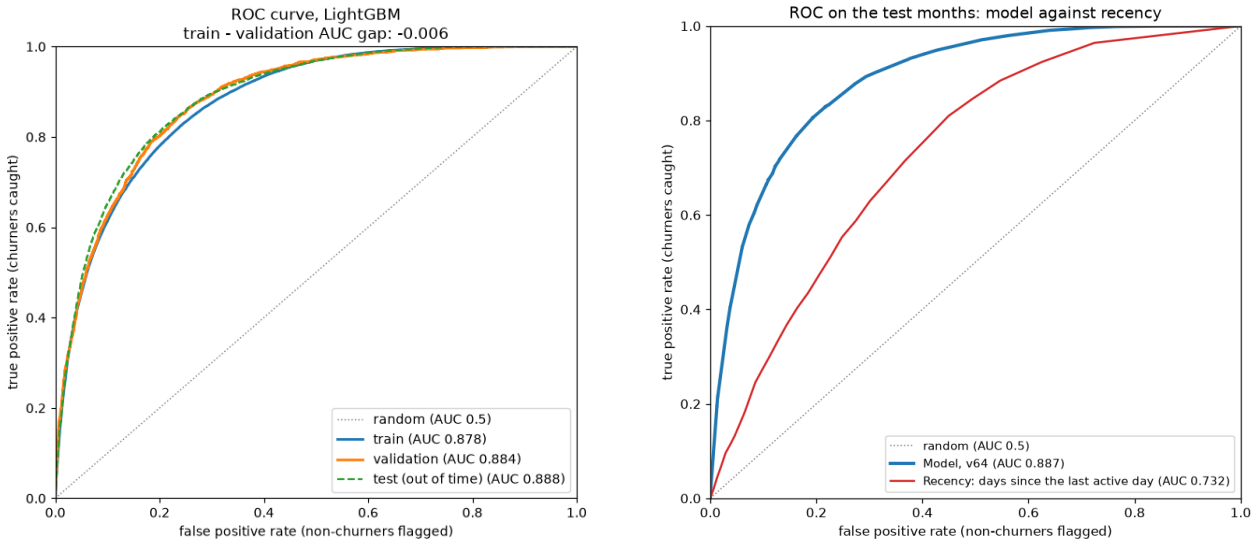


Figure 4: ROC curves of the registered model of brand 64 (`v64`). Left: training, validation and test. Right: the model against recency on the test months (July and August 2026).

Features. Figure 5 compares, for each family, how much the registered model uses it (share of the mean |SHAP|) with what it adds: the AUC lost on the test months when a LightGBM with all the features is trained again without the family (`docs/feature_ablation.md`, `make ablation`). Frequency weighs most (37% to 55% of SHAP), but without it the AUC only falls by 0.001 to 0.002, because recency, tenure and the lag and mix features carry the same activity information. **Tenure is the family the others cannot replace**: 0.007 to 0.013 of AUC and a worse log-loss in every brand. Monetary, trend, lag and deposit features add almost nothing (under 0.001 of AUC), and without any single family the GGR at risk reached by 1,000 contacts changes by 2 points at most.

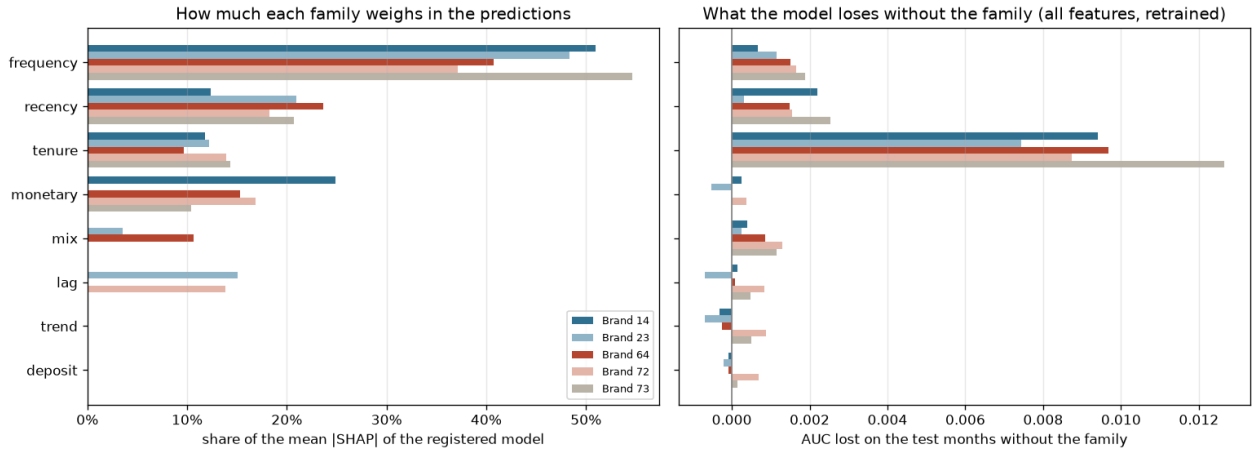


Figure 5: Importance by feature family. Left: share of the mean |SHAP| of the registered LightGBM of each brand on the validation month (a family with no bar is not in that model). Right: AUC lost on the test months when a LightGBM with all the features is trained again without the family. Source: `docs/feature_importance_brand*.md` and `docs/feature_ablation.md`.

Calibration. Raw LightGBM predicts too much churn, because `scale_pos_weight` gives more weight to the churners. Training tries isotonic and Platt and keeps the lowest cross-validated ECE on the validation month (Figure 6). For brand 64 they tie (ECE 0.006) and isotonic is kept. In test both have an ECE of 0.031: the probabilities are about 3 points too high in July and August (31.9% against a real 28.8%), although churn barely changed from June (28.6%). The level shift of Section 4.5 corrects this kind of gap on the scoring date. The test is only reported; it is never used to choose.

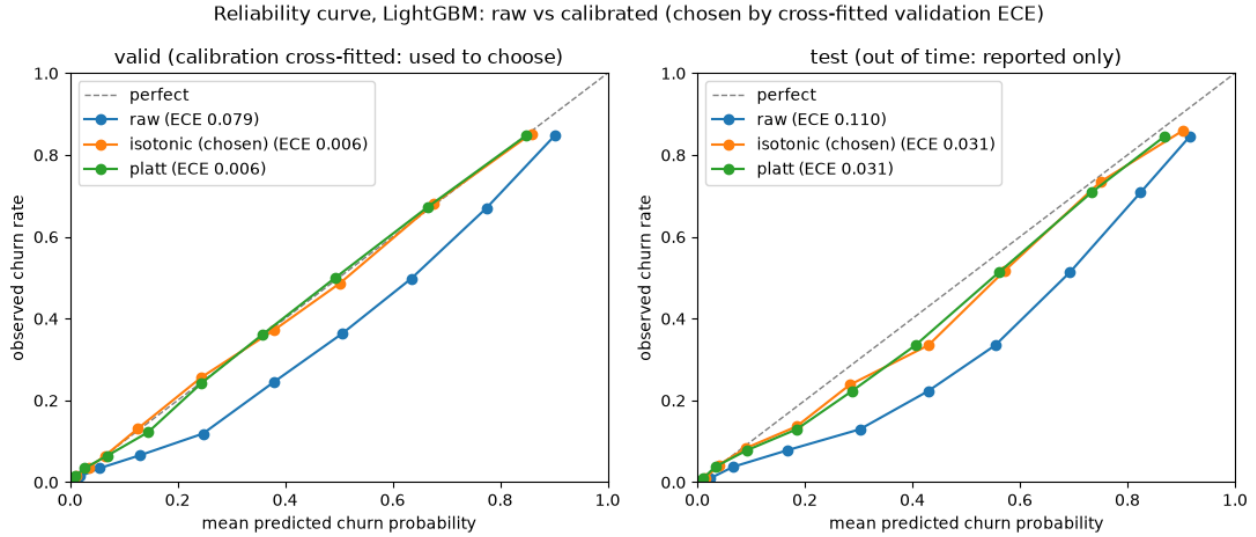


Figure 6: Calibration curves of the registered model of brand 64 (`v64`), as saved in MLflow: raw, isotonic (chosen) and Platt. Left: validation; right: test.

4.2 Backtest

Table 9: Backtest by brand: means over the 11 monthly test cutoffs, with 95% intervals.

Brand	Model AUC	Recency AUC	Cox C-index	Ratio R	Max ECE	GGR covered	Verdict
14	0.887 [0.883, 0.890]	0.775 [0.769, 0.782]	0.820 [0.799, 0.843]	1.35 [1.30, 1.41]	0.094	80% (28%) [77%, 82%]	FAIL
23	0.886 [0.877, 0.894]	0.785 [0.778, 0.792]	0.802 [0.779, 0.829]	1.39 [1.32, 1.46]	0.101	61% (10%) [57%, 65%]	FAIL
64	0.869 [0.858, 0.880]	0.759 [0.746, 0.771]	0.795 [0.773, 0.821]	1.28 [1.23, 1.35]	0.118	68% (19%) [61%, 73%]	FAIL
72	0.876 [0.868, 0.884]	0.760 [0.752, 0.768]	0.824 [0.811, 0.840]	1.19 [1.15, 1.23]	0.102	68% (14%) [63%, 71%]	FAIL
73	0.887 [0.880, 0.894]	0.779 [0.775, 0.782]	0.820 [0.806, 0.836]	1.25 [1.20, 1.29]	0.073	59% (8%) [56%, 62%]	PASS

Note. Values in brackets are 95% intervals of the mean from a circular block bootstrap: 2,000 resamples of blocks of 3 consecutive months, because neighbouring months move together. AUC: area under the ROC curve. C-index: concordance index of the Cox PH model. Ratio R: top-decile ratio, Equation (7). Max ECE: highest monthly expected calibration error, Equation (4). GGR covered: share of the GGR at risk covered by the 1,000 players with the highest value at risk, Equation (6); recency in round brackets. Red: condition failed. Source: `docs/backtest_v0_brand*.md`.

The model beats recency at every cutoff, by at least 0.078 of AUC, and its AUC is never below 0.83; Cox PH ranks almost as well (AUC 0.86 to 0.88). **Uncertainty**: the 11 months are not independent, so the intervals resample blocks of consecutive months, and Table 10 adds the median and the worst month, which need no resampling. The model beats recency in AUC and passes the 1.10x ratio in all 55 brand-months; its worst month is December 2025 in brand 64 (AUC 0.831) and its lowest ratio is 1.11 (brand 72, December 2025). Of the riskiest 10% of players, 80% to 94% really churn with the model, against 57% to 78% with recency. In brand 64 the riskiest 10%, 20% and 30% hold 29%, 54% and 72% of all churners, against 19%, 36% and 50% with recency (Figure 7). With the same 1,000 contacts, recency reaches 8% to 28% of the GGR at risk and a random choice 4% to 15%. The only failures are ECE failures (Section 4.5). The model also gives what recency cannot: calibrated probabilities, the 7 to 60-day horizons, risk bands and the drivers of each player.

Table 10: Backtest month by month: median and worst month of the model, with no resampling.

	Model AUC		Ratio R against recency	
Brand	median	worst month	median	worst month
14	0.888	0.877 (Feb 2026)	1.34	1.22 (Nov 2025)
23	0.892	0.863 (Apr 2026)	1.42	1.25 (Dec 2025)
64	0.871	0.831 (Dec 2025)	1.27	1.16 (Oct 2025)
72	0.873	0.857 (Oct 2025)	1.18	1.11 (Dec 2025)
73	0.885	0.867 (Dec 2025)	1.24	1.15 (Dec 2025)

4.3 Business value

Ranking by value at risk uses money and recency alone does not, so comparing the two is not fair. Each risk score therefore ranks the same players (Table 11), with the same 1,000 contacts, in two ways: by its probability alone (*value-blind*) and by $G_i\, p_i$ (*value-aware*). Cox at 30 days times value is the `value_at_risk_30d` that the daily scores handed off until 16 October 2026. A random score times value shows what value alone gives.

- **Value does most of the work.** Weighting LightGBM by value adds 58 to 76 points. The final policy reaches 49 to 54 points more than value-blind recency: 3 to 8 times more historical GGR opportunity.
- **Value alone already goes far.** A random score times value reaches 28% to 66% (Figure 8); LightGBM times value adds 3 to 31 points to it, most in the brands with less churn.

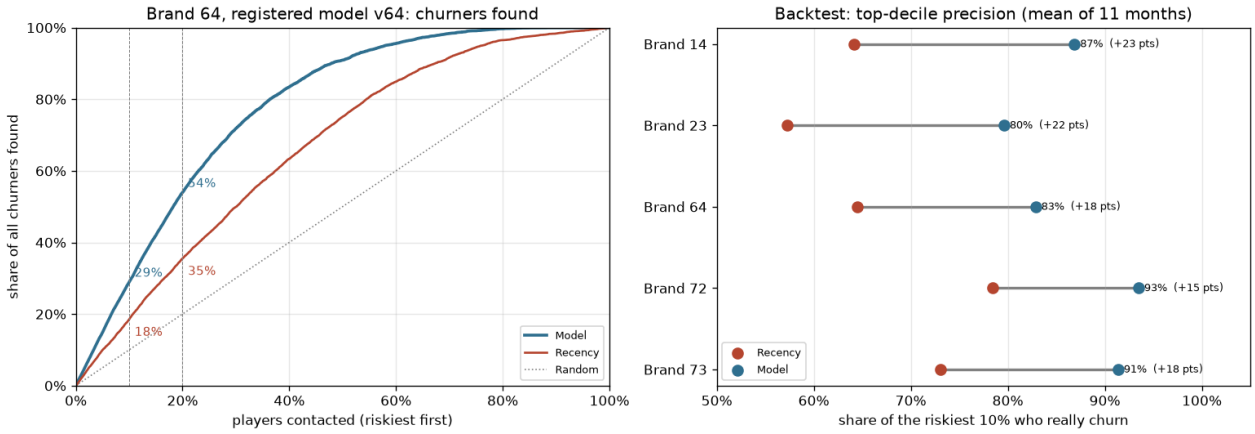


Figure 7: Model against recency. Left: share of all churners found when the players are contacted from the riskiest down (brand 64, registered model `v64`; test months July and August 2026; ties in recency broken at random). Right: mean top-decile precision over the 11 backtest months, for each brand, with the official active day. Source: MLflow run of `v64` and `docs/backtest_v0_brand*.md`.

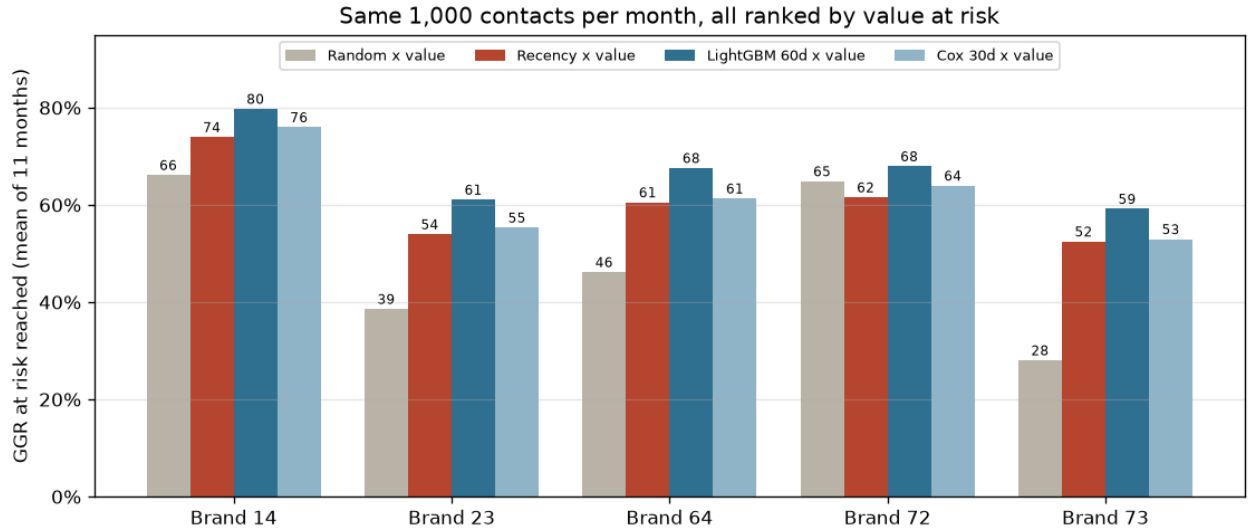


Figure 8: GGR at risk reached with the same 1,000 contacts per month, mean of the 11 backtest months: every score is weighted by value: a random score, recency, LightGBM at 60 days and Cox at 30 days. Source: backtest policy files.

Table 11: GGR at risk reached by each policy with the same 1,000 contacts, mean of the 11 backtest months, in %.

	Recency		LightGBM 60d		Cox 30d	Random	LightGBM × value better than
Brand	value-blind	× value	value-blind	× value	× value	× value	recency × value
14	31.0	73.9	3.8	**79.8**	76.0	66.2	10 of 11 months
23	10.1	54.0	2.3	**61.1**	55.3	38.6	11 of 11 months
64	18.3	60.5	4.8	**67.6**	61.3	46.2	11 of 11 months
72	14.4	61.6	4.5	**68.0**	64.0	64.9	11 of 11 months
73	7.2	52.5	1.4	**59.2**	53.0	28.1	11 of 11 months

Note. Value-aware policies rank by $G_i\, p_i$ (Equation (6)). Recency is the calibrated recency probability, so its ties differ a little from Table 9. From the columns: predictive uplift = LightGBM × value minus recency × value; value-weighting uplift = LightGBM × value minus LightGBM value-blind; combined = LightGBM × value minus value-blind recency. Source: `data/03_output/backtest/*_policies.csv`.

- **The model adds a steady extra.** Against recency, both weighted by value, LightGBM adds 5.9 to 7.1 points and is better in 54 of the 55 months, €147k to €473k per brand over 11 months. Recency weighted by value already reaches 88% to 93% of the final policy.
- **The handoff score is the weaker one.** In the backtest, Cox at 30 days times value reaches 3.8 to 6.3 points less than LightGBM at 60 days times value in every brand (€103k to €437k over 11 months), and the two top-1,000 lists share only 63% to 68% of the players. Part of this gap is not fair: in the backtest the Cox horizon probabilities are not calibrated (Section 4.4), while LightGBM is. With both models calibrated, on the test months of the registered models (July and August 2026), LightGBM still reaches more in all 10 brand-months, by 0.4 to 8.7 points (3.5 on average). So since 16 October 2026 the value at risk handed off (`value_at_risk`) uses the LightGBM 60-day probability.
- **Opportunity, not money saved.** The money saved is the GGR reached times the share of contacted churners that the campaign keeps, and this share needs a campaign with a control group.

4.4 Cox PH per horizon

The daily scores also give the chance that churn starts within 7, 14 and 30 days. Table 12 checks each horizon against its own label (`churn_within_7d/14d/30d`) on the players whose label is known. Cox ranks well (AUC 0.84 to 0.88). Its ECE in the backtest, 0.09 to 0.15, is the ECE of **uncalibrated** probabilities: on each cutoff the validation month is about 61 days old, and a label of churn within h days needs $h + 60$ days to close, so no horizon calibrator can be fitted point-in-time and the raw probability is used. In the registered models, whose validation month has closed labels, the calibrated 30-day ECE on the test month is 0.010 to 0.025. I compare Cox with LightGBM only with the same metric at the same horizon, because the C-index measures the order over the whole follow-up.

Table 12: Cox PH per horizon: AUC and ECE against the label of each horizon, mean of the 11 backtest months. The probabilities are not calibrated (see the text).

	AUC			ECE		
Brand	7 days	14 days	30 days	7 days	14 days	30 days
14	0.880	0.877	0.873	0.107	0.101	0.094
23	0.866	0.858	0.848	0.096	0.094	0.092
64	0.853	0.847	0.836	0.104	0.097	0.093
72	0.868	0.867	0.867	0.150	0.135	0.114
73	0.874	0.870	0.866	0.114	0.104	0.088

Proportional hazards (`docs/cox_ph_check.md`, `make ph-check`). Cox PH assumes that each covariate multiplies the hazard by the same factor at every time. With 42,000 to 191,000 events per brand the Schoenfeld test rejects it for almost every covariate, so I measure how much each effect changes: the coefficient on the events of each time window, $\beta(t)$, is β plus the mean scaled Schoenfeld residual of those events (Grambsch and Therneau, 1994) (Figure 9). Frequency keeps its sign and is 1.2 to 2 times stronger at the start than after 90 days. Recency raises the hazard of churn on the cutoff day (hazard ratio 1.18 to 1.32 per unit), lowers it in days 1 to 7 in four brands and has almost no effect after day 30. Prior dormancy spells are strong on day 0 and weak after 90 days. So the assumption does not hold for the strongest covariates: Cox averages effects that belong to different times. **It can be fixed, but the fix changes nothing in practice.** Censoring the follow-up at 60 days and stratifying by recency group (0, 1 to 7 and over 7 days) removes every sign change in the five brands, but the horizon AUC (0.863 against 0.864 at 30 days) and the calibrated ECE (0.019 against 0.017) stay the same. So I keep Cox as it is, to rank the players by horizon.

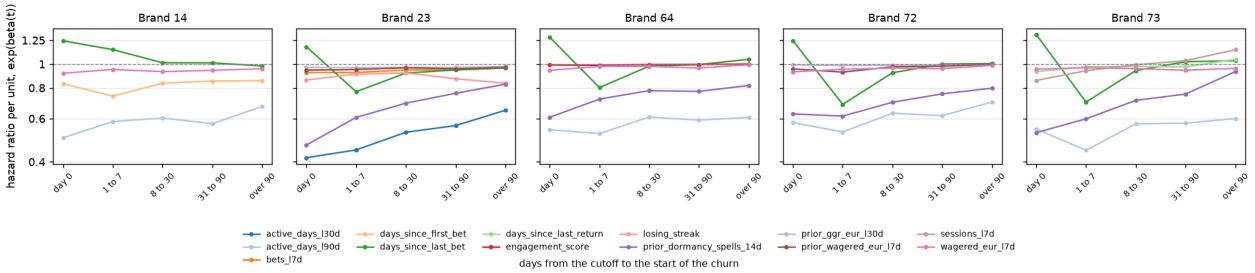


Figure 9: Effect of each Cox PH covariate over time, $\exp(\beta(t))$, by window of days from the cutoff to the start of the churn (training rows of the registered model of each brand). A flat line is a proportional effect. Source: `docs/cox_ph_check.md`.

4.5 Seasonal calibration

The calibration month is 2 to 3 months older than the scoring date, because the label needs 60 days, and the churn rate moves a lot from month to month (Figure 10, top): 26% to 42% in brand 14, 17% to 40% in brand 23, 23% to 48% in brand 64, 42% to 66% in brand 72 and 33% to 52% in brand 73. The ranking stays right, but the level of the probability does not. On the scoring date `src/models/calibration_level.py` shifts all the probabilities p_i of brand b in log-odds:

$$p_i' = \sigma\big(\mathrm{logit}(p_i) + \delta_b\big), \qquad \delta_b = \mathrm{median}_{m \in M}\ \delta_{b,m}, \qquad |\delta_b| \le 2, \tag{8}$$

where σ is the logistic function, M holds the last 3 closed months and the recent months with at least 14 days of data, and $\delta_{b,m}$ makes the mean shifted probability of month m equal to its churn rate. The shift is the same for every player, so the AUC and the top decile do not change.

Recent months. A recent cohort c has only $k = D - c \ge 14$ days of data on the scoring date D, so its 60-day churn rate is estimated with a **partial-label correction**:

$$\hat{r}_c = s_c(k)\,\frac{S_{\mathrm{ref}}(60)}{S_{\mathrm{ref}}(k)}, \tag{9}$$

where $s_c(k)$ is the share of cohort c with no active day in its first k days, and $S_{\mathrm{ref}}(k)$ and $S_{\mathrm{ref}}(60)$ count the players of the last cohort with a complete label on D that had no active day after k and after 60 days. The fraction is the chance that a player silent for k days stays silent for 60. It uses activity up to D only and no proxy outcome (`calibration_level.early_rate`).

Point-in-time check. I replayed the estimate for every monthly cohort of the five brands 14, 21, 30 and 45 days after its cutoff, each time with the data up to that day only (Table 13), and an automated test checks that the reference cohort is always complete on the replay day. After 14 days the estimate is 2.2 points from the real rate, against 5.3 for the last closed month, and it is closer in 208 of the 250 replays. This error is small next to monthly changes of 16 to 25 points, and the median over several months softens it further. One limit: gold was rebuilt on 8 August 2026, so the replay is point-in-time in the days it reads, but not in later corrections of those days.

Result. The ECE peaks match the jumps in churn. With the level shift, the months above 0.08 go down from 11 to 5: December 2025 in brands 14, 23, 64 and 72, and February 2026 in brand 64; brand 73 has none. In these months the change happens after the scoring date, so no earlier data can show it. One more year of history will allow a year-on-year seasonal term.

4.6 Top-decile condition

The plan asked for a top-decile precision of 1.5 times recency. Recency is already 57% to 78% precise, so even a perfect model cannot pass 1.27x to 1.75x on average, and the ceiling is below 1.5x in 33 of the 55 brand-months (Figure 11); the model reaches 1.5x in only 3 months. **I propose 1.10x**: on 9 October 2026, with the first backtests, it was the strictest round value that every brand passed with 95% confidence. Now the lower ends of the intervals are 1.15 to 1.32, every month passes 1.10x, and I do not raise the limit after seeing these results (`docs/decision_top_decile_condition.md`).

Table 13: Early churn estimate of Equation (9) against the real 60-day churn rate. Mean absolute error over the cohorts, in percentage points (pp): 2.2 pp means an estimate of 37.8% or 42.2% for a real rate of 40%.

			After 14 days		After 30 days		
Brand	Cohorts	Mean real churn rate	Estimate (pp)	Last closed month (pp)	Estimate (pp)	Last closed month (pp)	Estimate closer
14	12	34.1%	2.2	5.2	1.3	3.6	33 of 50
23	12	25.3%	2.7	5.6	1.1	3.8	44 of 50
64	12	32.8%	2.3	6.1	1.1	5.1	42 of 50
72	12	51.8%	2.4	5.7	1.2	5.3	45 of 50
73	12	40.6%	1.6	4.1	0.9	3.7	44 of 50
All	60	36.9%	2.2	5.3	1.1	4.3	208 of 250

Note. Cohorts: monthly cohorts replayed after 14 days (13 after 30 days). Estimate closer: replays at 14, 21, 30 and 45 days where the estimate is closer to the real rate than the last closed month. Source: backtest early-rate files.

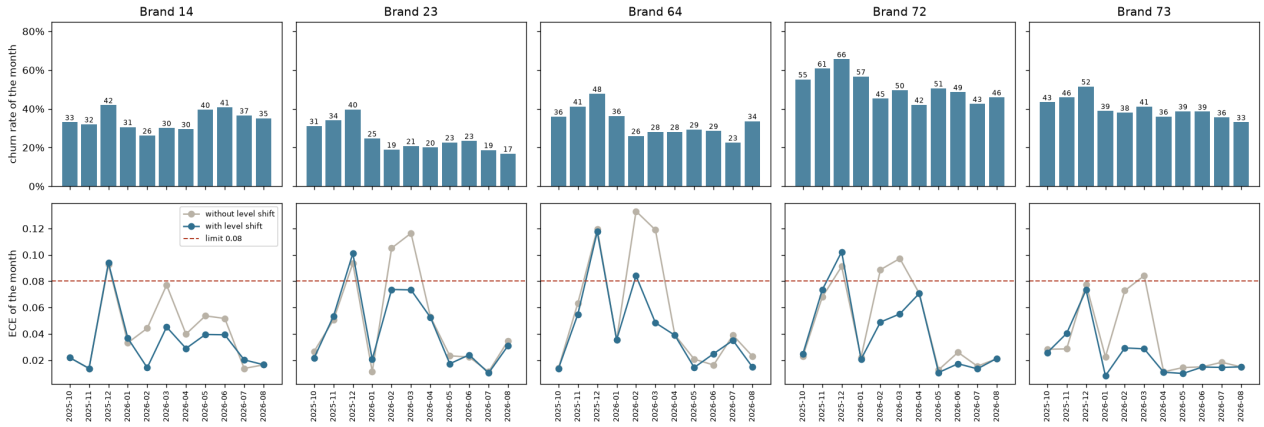


Figure 10: Backtest month by month. Top: churn rate of each test month. Bottom: ECE without the level shift (grey) and with it (blue); red line, limit of 0.08.

Freezing the limit. Backtests on months already seen cannot confirm a limit chosen after seeing them, and this includes the reruns after the active days rebuild. So 1.10x must be approved before the first unseen test month is read: September 2026, whose label closes on 31 October 2026.

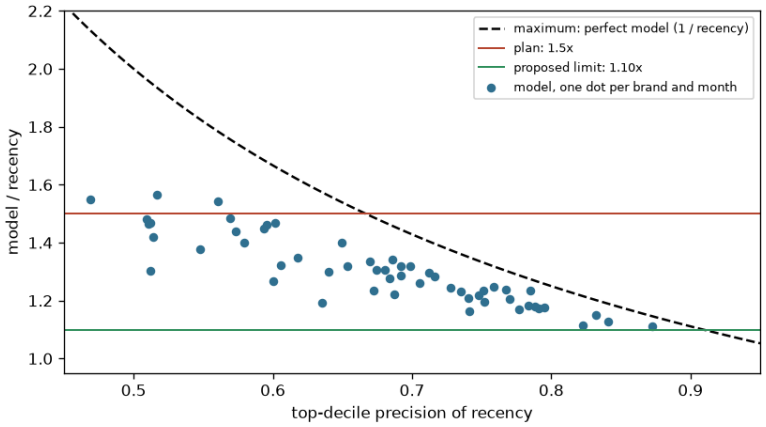


Figure 11: Each dot is one brand and one month. The dashed line is the highest possible value; in 33 of the 55 brand-months it is below 1.5x.

5 Validation of the signals table

I checked the 28 columns of `gld_player_signals_daily` that the features read, in five brands (`eda/signal_validation.py`, report in `docs/signal_validation.md`). 21 columns are recomputed from the daily tables, and a column is right if it matches (within 2% or 0.5) for 95% of the players. The same recomputation 7 days later detects leaks, and two leaks planted on purpose are caught. The other columns are checked for valid ranges, documented formulas and values. **18 of the 28 columns pass every check** (Figure 12); Table 14 lists the problems.

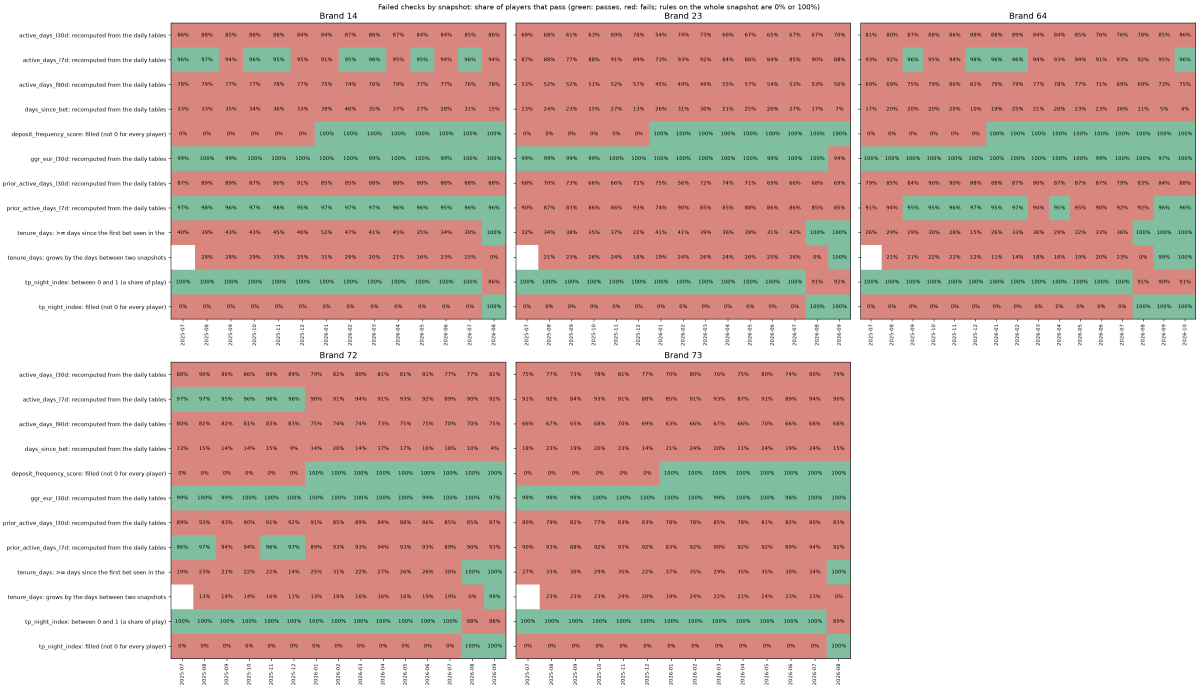


Figure 12: Failed checks by brand and monthly snapshot (green passes, red fails).

Table 14: Problems in the signals table and their effect.

Column	Problem	Effect
`days_since_bet`	0 for almost everyone until July 2026; from August, the recency of 7 days later	Not used; computed again
`tenure_days`	Wrong in the history; right from September 2026	Not used; computed again
`active_days_*`	Counts the days with stakes only from playable bonus: `active_days_l90d` matches the official count for only 34% to 54% of the players, and it is never lower	Not used; computed again from the active days (Section 2.2)
`tp_night_index`	Empty until August 2026; goes up to 4, so it is not a share	No model uses it
`deposit_frequency_score`	Empty before 2026	No model uses it
`ggr_eur_l30d`	Just below the limit in one snapshot (brand 23, September 2026: 94% of the players match)	No model uses it

6 Plan versus delivery

Table 15 gives the totals and Table 16 every Phase 1 item of the Week 1 delivery plan with one status: **Done** (as the plan asks), **Changed** (in a different way, for the reason given), **Deferred** (moved to a later phase by an outside constraint) or **Not done** (not delivered yet). *Ref.* is the plan reference:

a task (T1 to T12), a deliverable (D1 to D9) or a framework section.

Table 15: Status of the 40 Phase 1 items.

Status	Items	Main reasons
Done	25	
Changed	14	60-day churn; calibration choices; 11 cutoffs; KPI threshold; value at risk at 60 days; local MLflow and Feast
Deferred	1	Read-only AWS access in the sprint
Not done	0	None

Table 16: Plan versus delivery for Phase 1.

#	Plan item	Ref.	Status	What I delivered, or why it differs	Evidence
A. Repository and tracking (T1, D1)					
1	Pinned environment, pre-commit, `make setup`, tests	T1	**Done**	uv lockfile, ruff and black in pre-commit, `make setup` and `make test`.	`uv.lock`, `Makefile`
2	Folders `data/`, `eda/`, `features/`, `models/`, `evaluation/`, `serving/`	T1	**Changed**	The same names, with the Python packages under `src/` (`src/features`, `src/models`, `src/evaluation`).	`src/`
3	MLflow tracking server on EC2 with S3	T1	**Changed**	Local MLflow (SQLite and local artefacts). Every run is logged with its config. A remote server is Phase 2 infrastructure.	`mlflow.db`, `make mlflow-ui`
B. Data, EDA and churn definition (T2 to T4, D2, D3)					
4	Access and register of the source tables	T2	**Changed**	I used the gold layer of the S3 lake (`org/40-gold`): signals, financial and payments tables. The raw bet, session and bonus tables are not exposed in the sprint.	`docs/data_card.md`, `eda/01_profiling.ipynb`
5	EDA stage 1: data-quality suite per source table and per feature snapshot	EDA	**Done**	pandera suites (the plan allows Great Expectations or pandera) and one report per table.	`eda/expectations/`, `docs/dq_reports/`
6	EDA stage 2: outliers and anomalies; winsorisation config	EDA	**Done**	IQR, robust z-score, Isolation Forest, daily totals and impossible sequences. Caps at p99.5, learned on the training months.	`eda/anomalies.py`, `configs/winsorisation_*.yaml`
7	EDA stage 3: alerts with thresholds; a critical alert stops the run	EDA	**Done**	The same thresholds are reused when scoring.	`configs/eda_alerts.yaml`, `src/features/alerts.py`
8	EDA stage 4: summary against the benchmark	EDA	**Done**	Run for brands 23, 64 and 72.	`eda/summary.py`, `docs/eda_summary_brand*.md`
9	Churn definition v0: KM curves for 14, 30 and 60 days, return rates	T3, D2	**Changed**	60 days of silence instead of 30: with 30 days, 21% to 29% of the churned players come back. Active day as defined in the gold data model (Section 2.2).	`docs/churn_definition_v0.md`
10	Churn signal explorer: promoted and rejected signals	T4, D3	**Done**	32 hypotheses tested and recorded.	`eda/signal_explorer.py`, `docs/churn_signals_v0.md`
C. Features (T5, T6, D4)					
11	Feature dictionary with at least 25 features	T5, D4	**Done**	39 features, with source, window and leakage note.	`docs/feature_dictionary.md`

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Table 16, continued

#	Plan item	Ref.	Status	What I delivered, or why it differs	Evidence
12	`build_features --as-of`, 12 monthly cutoffs, leakage test	T6, D4	**Done**	Point-in-time features and a pytest that fails on leakage.	`src/features/`
13	Feast registry and offline store	T6, D4	**Changed**	Local registry and offline store, with a point-in-time check. S3 needs write access.	`src/features/feature_repo/`
14	Versioned Parquet snapshot in S3	T6	**Deferred**	AWS access is read-only in the sprint. The snapshot is versioned in MLflow instead.	MLflow artefacts
15	Feature selection: missing values, variance, correlation, adversarial validation, permutation pruning	Features	**Done**	Applied on the training months only.	`src/models/select_features.py`
16	Feature importance: permutation (with CIs) and SHAP	Features	**Done**	One report per brand.	`docs/feature_importance_brand*.md`
17	Feature importance: ablation by feature family	Features	**Done**	A LightGBM with all the features is retrained without each family; AUC, log-loss, ECE and GGR reached are reported (Figure 5).	`docs/feature_ablation.md`
D. Survival dataset (T7, D5)					
18	Duration, event, right censoring; 7, 14 and 30-day labels	T7, D5	**Done**	Built per cutoff from the 60-day definition.	`src/features/build_survival_dataset.py`
19	Censoring rate and duration distribution	T7, D5	**Done**	Reported per cutoff.	`docs/eda_summary_brand64.md`
20	Left truncation handled and explicit; survival data card	T7, D5	**Done**	No left truncation: time starts at the cutoff and every player enters on day 0. Censoring rate, durations and a check against the 60-day label in the card.	`docs/survival_data_card.md`
E. Evaluation and models (T8, T9, D6, D7)					
21	Evaluation module: C-index, time-dependent AUC, IBS, one-calibration, ECE, top-decile precision, bootstrap CIs; `evaluate --run-id`	T8, D6	**Done**	With unit tests on synthetic data.	`src/evaluation/`
22	Reference LightGBM and Cox PH runs; results table	T9, D7	**Done**	Results table and one model card per brand.	`docs/results_v0_brand64.md`, model cards
23	Served score: LightGBM `p_churn_30d`	Model	**Changed**	`p_churn_60d`, which follows from the 60-day definition.	`docs/schema_player_scores.md`
24	Calibration: isotonic or Platt, chosen by ECE on the test months	Optim.	**Changed**	Chosen by cross-fitted ECE on the validation month, so the test months are not used to decide.	`src/models/calibration.py`
25	Calibration level re-estimated on the scoring date (not in the plan)	Added	**Changed**	The calibration month is 2 to 3 months old and the churn rate is seasonal. It uses data up to the scoring date only; a replay at 14 to 45 days checks it (Table 13).	`src/models/calibration_level.py`
26	Optimisation steps: defaults, class imbalance, Optuna, regularisation, pruning, temporal robustness, inference cost	Optim.	**Done**	Tuning adds nothing over the defaults, so I keep the simpler reference configuration.	`src/models/train.py`, `docs/temporal_robustness_brand64.md`

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Table 16, continued

#	Plan item	Ref.	Status	What I delivered, or why it differs	Evidence
27	`train --config configs/lightgbm.yaml`	Training	**Changed**	`train --model-id` with a model catalog; the full config is logged with each run.	`src/models/train.py`
28	Cox PH diagnostics per horizon: time-dependent AUC and Brier at 7, 14 and 30 days; proportional hazards	Model	**Done**	AUC, Brier and ECE per horizon are in the backtest (uncalibrated there; calibrated 30-day ECE 0.010 to 0.025 in the registered models). Proportional hazards does not hold for recency and prior dormancy; a fix with censoring and strata changes neither AUC nor calibration, so Cox is kept (Section 4.4).	`docs/cox_ph_check.md`
F. Backtest (T10, D8)					
29	Rolling-origin backtest over 12 monthly cutoffs	T10, D8	**Changed**	11 test months: the data start in April 2025, the features need 90 days and each test month needs two earlier months with a closed 60-day label. A backtest run from 31 October 2026 has 12, with no code change (Section 3.5). Weekly scoring days not checked yet.	`docs/backtest_v0_brand*.md`
30	Metrics per cutoff; comparators recency, reference LightGBM and Cox PH	T10	**Done**	AUC, C-index, log-loss, ECE, top-decile precision and recall, Brier. Intervals by block bootstrap of consecutive months, with the median and the worst month.	`src/models/backtest.py`
31	Comparator: incumbent churn buckets	T10	**Changed**	The source `churn_score` is computed but left out of the report: it uses a 30-day definition.	`src/models/backtest.py`
32	Pass condition: AUC never below 0.70	T10	**Done**	Met at every cutoff in all five brands.	`docs/backtest_v0_brand*.md`
33	Pass condition: top-decile precision at least 1.5 times recency	T10	**Changed**	Lowered to 1.10 times: 1.5 times is out of reach (the ceiling is 1.24 to 1.46 times). Proposed; it must be approved before the first unseen month (September 2026, label closed on 31 October 2026) is evaluated.	`docs/decision_top_decile_condition.md`
34	Pass condition: ECE at most 0.08 every month	T10	**Done**	Reported. Not met in 5 brand-months: December 2025 in brands 14, 23, 64 and 72, and February 2026 in brand 64.	`docs/backtest_v0_brand*.md`
35	Targeting replay at the agreed N, ranked by `value_at_risk_30d`	T10, D8	**Changed**	Ranked by LightGBM 60-day probability times expected value, the score handed off since 16 October 2026 (`value_at_risk`). Value-matched baselines and the Cox 30-day policy are replayed with the same contacts (Section 4.3). N = 1,000 is not yet agreed with CRM.	`docs/backtest_v0_brand*.md`
36	`backtest --config` and the backtest report	T10, D8	**Done**	One report per brand.	`configs/backtest.yaml`

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Table 16, continued

#	Plan item	Ref.	Status	What I delivered, or why it differs	Evidence
G. Scoring and end to end (T11, T12, D1, D9)					
37	`score --as-of --model-version` writing `player_scores`	T11, D9	**Done**	One row per active player; schema documented.	`src/models/score.py`, `docs/schema_player_scores.md`
38	`value_at_risk_30d` = expected 30-day GGR times P(churn within 30 days) from the served LightGBM	Design	**Changed**	`value_at_risk` = expected 30-day GGR times the 60-day probability of the served LightGBM, because the LightGBM score is at 60 days and it reaches more GGR at risk than the Cox 30-day probability, also with both calibrated (Section 4.3).	`docs/schema_player_scores.md`
39	Reserved columns `lower` and `upper`	Design	**Done**	Present and empty (prediction intervals are later research work).	`docs/schema_player_scores.md`
40	`make pipeline AS_OF=DATE` from a clean clone; demo; Week 1 summary	T12, D1	**Done**	Run from a clean clone on 2026-10-14 (11 minutes, plus the backtest).	`docs/week1_summary.md`

Out of scope for Phase 1. The plan also describes Airflow scheduling, EC2, ECS and Fargate deployment, FastAPI serving in the cloud, the Redis online store, a remote MLflow server, the production SLA, monitoring and the runbook. These belong to Phase 2 and are not counted as Phase 1 gaps. A local FastAPI service and its Docker image were started early (`serving/`), but they are not part of this assessment.

7 Conclusions and limitations

Conclusions

- **The model ranks churn risk well and steadily**: backtest AUC 0.87 to 0.89, 0.10 to 0.12 above recency and never below 0.83 in a month. Tuning adds nothing over the reference configuration.
- **The money comes mostly from using value**: 3 to 8 times more historical GGR opportunity than value-blind recency, and the model adds 6 to 7 points over recency weighted by value. How much of it is kept needs a campaign with a control group.
- **The value at risk uses LightGBM at 60 days**: with both models calibrated, it reaches more GGR at risk than Cox at 30 days in every brand-month tested. Cox stays to rank the players by horizon; its lack of proportional hazards does not change its results.
- **The level depends on the season**: the only failures are calibration failures, and the level shift reduces them from 11 to 5 brand-months.
- **The signals table is not reliable for recency, tenure and active days**: the pipeline does not depend on it for these features.

Limitations

- 18 months of history, with no full previous year, and deposits only from March 2026.
- The backtest cutoffs are not independent. The intervals resample blocks of consecutive months but not players, so they can still be a little optimistic; the month-by-month results need no resampling.
- The campaign replay measures who is chosen, not the effect of the campaign.

- The feature store, the versioned snapshot and MLflow are local, because I only have read access to AWS.

A How to reproduce the project

Table 17 lists the main commands. The environment is pinned with `uv` (`pyproject.toml`, `uv.lock`, Python 3.13). `make test` passes 24 project tests and 23 tests of `eda/dq_lib`. MLflow is local (SQLite).

Table 17: Main commands.

Command	What it does
`make pipeline AS_OF=... [BRAND=...] [BACKTEST=1]`	The full pipeline on one date
`make backtest AS_OF=... BRAND=...`	11-month backtest
`make score AS_OF=... [MODEL_VERSION=n]`	Scores of the day
`make model-card BRAND=...`	HTML model card
`make validate-signals`	Validation of the signals table
`make test` / `make mlflow-ui`	Tests / MLflow interface

B Abbreviations and terms

Active day
A day with a stake from real money or released bonus funds; playable bonus alone does not count, Equation (1).

AUC
Area under the ROC curve: probability that the model gives more risk to a player who leaves than to one who stays; 0.5 is random.

CI
Confidence interval (here, 95% bootstrap interval).

CRM
Customer relationship management team, who would contact the players.

Calibration
Turning the model score into a real probability.

Censoring
The churn cannot be seen yet because the data ends before.

C-index
The AUC of survival models.

Cox PH
Survival model: it estimates when churn comes.

ECE
Expected calibration error: mean difference between the predicted probability and the real rate, Equation (4).

GGR
Gross gaming revenue: money wagered minus money paid in wins.

IBS
Integrated Brier score: mean error of the survival curve; lower is better.

IQR, MAD
Interquartile range and median absolute deviation, two robust measures of spread.

KPI
Key performance indicator.

Log-odds
Logarithm of $p/(1-p)$; shifting it moves all probabilities without changing their order.

Point in time
Using only the data available on the cutoff date.

PSI
Population stability index: how much the distribution of a feature has changed.

ROC
Receiver operating characteristic curve: true positive rate against false positive rate.

Risk band
Decile of the churn probability within the brand and day.

SHAP
Shapley additive explanations: they split each player's score between the features.

Top decile
The 10% of players with the highest risk.

Value at risk
Expected GGR in 30 days times the churn probability, Equation (6).

Winsorisation
Capping extreme values at a percentile instead of deleting them.

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